

The Long Shadow of Loyalty: Political Stigma and Bureaucratic Careers in Ming China

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April 30, 2026

[Preliminary draft]

Abstract

How do regime transitions shape the careers of later elites from places associated with loyalty to a defeated ruler? We study this question through the Jingnan Rebellion (1399–1402), when the Yongle emperor seized the Ming throne while preserving the examination-based recruitment system. Linking examination records to career histories for *jinshi* in Ming China, we show that the political penalty for association with Jianwen loyalist martyrs operated less at examination entry than during subsequent bureaucratic careers. Post-1402 *jinshi* from martyr counties did not suffer a systematic decline in top examination placement, but reached lower peak ranks, were more likely to remain in low-ranking offices, and were less likely to serve in the central government. The penalty extended beyond identifiable martyr lineages and instead attached to politically marked localities. We interpret this pattern as displaced political screening: meritocratic entry was preserved, while political trust was reallocated at later, less visible stages of career advancement. The findings show how regimes can consolidate power through personnel allocation within inherited institutions rather than visible institutional rupture.

Keywords: Political Loyalty, Collective Stigma, Political Screening, Bureaucracy.

JEL Codes: D73, D74, H11, J45, N45

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1 Introduction

Do the political costs of loyalty to the defeated side extend to later elites from the same places? For rulers who come to power through contested succession, a central problem is that association with the defeated side may remain politically consequential even after the conflict has ended. Yet such rulers still depend on a functioning bureaucracy, making loyalty concerns difficult to manage. The key question is whether those costs persist for later officials from politically associated places. More broadly, this problem speaks to a central political economy question: how rulers consolidate power when they inherit institutions and personnel they cannot fully trust (Egorov and Sonin, 2011; Gandhi and Przeworski, 2006; Francois et al., 2015; Myerson, 2008; Acemoglu and Robinson, 2001).

This paper studies that question in the aftermath of the Jingnan Rebellion of 1399 to 1402, the civil war through which the Yongle emperor seized the Ming throne from the Jianwen emperor. The episode is especially useful for this question because the conflict made political allegiance legible, while the Ming state’s core institutions, including the examination-based recruitment system, remained in place. Officials who died for the Jianwen regime became visible markers of local political association, allowing us to trace whether that association shaped the careers of later elites from the same counties. This combination of political rupture and institutional continuity allows us to distinguish between two stages of elite selection that are often conflated: formal entry through the *jinshi* examination, the main gateway into official careers, and subsequent career allocation within the bureaucracy.¹

We construct an individual-level dataset of Ming *jinshi* from 1371 to 1487 by combining examination records with subsequent career histories. Treatment is defined at the county level: a county is coded as treated if it produced a Jianwen loyalist martyr during the Jingnan conflict. Such counties were politically salient after 1402 because loyalty to the Jianwen emperor could mark a place as tied not merely to the losing side of a civil war, but to a reigning ruler whose overthrow raised acute questions of legitimacy. In that setting,

¹Not all conflicts are equally informative for this question. Under wholesale dynastic collapse, foreign conquest, or severe administrative breakdown, later career outcomes may reflect multiple overlapping channels, making political exposure harder to isolate and treatment harder to define. Relative to such cases, the Jingnan episode provides a narrower and more tractable county-level source of political exposure under continued bureaucratic institutions.

local association with Jianwen loyalist martyrs could make later elites from those places politically suspect under the Yongle regime. We then compare how outcomes changed after 1402 for *jinshi* from martyr and non-martyr counties, using county and examination-year fixed effects together with a range of tighter comparison designs.

Our central finding is that the post-1402 disadvantage emerged after entry into the bureaucracy rather than at the examination stage itself. We find little evidence that candidates from martyr counties were systematically screened out of top examination placement after the regime change. Conditional on becoming *jinshi*, however, they faced substantial career penalties in subsequent assignments and promotions. Our baseline estimates show materially weaker lifetime advancement for post-1402 *jinshi* from martyr counties. Their average peak rank was 1.039 points lower on the 18-point scale, equivalent to 13% of the pre-1402 martyr-group mean. To put this in administrative terms, the pre-period average for martyr-county *jinshi* corresponded roughly to a mid-fifth-rank position—comparable to a vice prefect or a mid-level ministry official. The estimated penalty would shift a typical official toward the sixth-rank level, closer to the ceiling faced by ordinary county magistrates. They were also 17.7 percentage points more likely to end their careers at rank 6 or below, indicating a greater likelihood of remaining in the lower and middle tiers of the bureaucracy, and 12.6 percentage points less likely ever to serve in the central government. In sum, the evidence points to a post-entry mechanism: distrust operated primarily through bureaucratic assignment and promotion rather than exclusion at the point of examination success.

Two additional results sharpen this interpretation. First, the disadvantage became more pronounced after 1421, when the capital moved from Nanjing to Beijing. The worsening is especially clear for rank-based outcomes. For access to central posts, part of the decline reflects a broader southern disadvantage after the move to Beijing, but officials from martyr counties still underperformed relative to other southern officials. Second, the evidence is more consistent with county-level political stigma than with narrow lineage-based retaliation. Post-1402 *jinshi* from martyr counties faced similar disadvantages whether or not they shared a surname with a recorded martyr, and the penalty also persists among counties with substantial pre-1402 *jinshi* pipelines. The disadvantage thus attached to politically marked places, rather than only to identifiable loyalist families or to weaker local elite reproduction.

Taken together, these patterns point to displaced political screening within a continuing bureaucracy: political information mattered less at the formal entry stage than in later assignments, promotions, and access to the center. The regime preserved the formal structure of merit-based recruitment while reallocating trust and advancement at later stages of official careers. More broadly, the findings show how regime consolidation can proceed through personnel allocation inside inherited institutions rather than through visible institutional rupture.

This paper contributes first to research on authoritarian personnel systems, bureaucratic control, and the loyalty–competence trade-off (Egorov and Sonin, 2011; Gandhi and Przeworski, 2006; Francois et al., 2015; Iyer and Mani, 2012; Colonnelli et al., 2020; Akhtari et al., 2022; Jiang, 2018; Jia, 2022; Wang et al., 2022). A central theme in this literature is that rulers value capable agents but fear politically unreliable ones. Recent work has shown that regimes may directly target cultural and intellectual elites perceived as threats (Esberg, 2025). Our evidence identifies a distinct organizational margin: political screening need not operate through overt exclusion or direct repression, but can instead work quietly after recruitment, through postings, promotion, and access to the center. The regime preserved the visible structure of merit-based entry while reallocating career advancement away from politically marked groups. This perspective also distinguishes our paper from related work on loyalty choices and collaboration during conflict, which studies how peer influence, social networks, and career incentives shape individual decisions to defect or remain loyal (Cagé et al., 2023; Guo et al., 2024; Fan et al., 2026). Rather than asking why individuals switch sides during conflict, we examine how a regime subsequently allocates trust and careers to later elites associated with the defeated side.

Second, the paper contributes to research on political connections and collective reputation by shifting attention from the benefits of proximity to power to the *costs* of association with a defeated political side. Existing work shows that ties to incumbents can improve access to rents, protection, and promotion (Fisman, 2001; Faccio, 2006). Our evidence shows that the same logic can run in reverse: association with a politically tainted locality can become a durable career liability after power changes hands. Importantly, the relevant marker was not family identity but county of origin. Later jinshi from martyr counties faced similar

disadvantages whether or not they shared a surname with a recorded loyalist. This connects to theories in which group-level markers shape individual opportunities even when personal quality is not directly observed (Spence, 1973; Tirole, 1996), and complements recent work on how political shocks reshape power across places over time (Bai et al., 2023; Chen et al., 2025). Relative to that literature, our evidence highlights a specific channel of persistence: place-based political stigma operating within an otherwise continuous bureaucratic system, rather than through institutional destruction or wholesale elite replacement.

The rest of the paper proceeds as follows. Section 2 provides historical background on the Jingnan conflict, the treatment definition, and the importance of the 1421 capital relocation. Section 3 describes the data and empirical strategy. Section 4 presents the main results, dynamic patterns, heterogeneity, and robustness checks. Section 5 concludes.

2 Context

2.1 The Jingnan Shock and Its Political Meaning

The conflict began after the death of the Hongwu emperor in 1398. His grandson, the Jianwen emperor, moved to weaken the military princes, and in 1399, Zhu Di, the Prince of Yan, launched a rebellion. In 1402, his forces entered Nanjing, Jianwen disappeared from the historical record, and Zhu Di became the Yongle emperor. Military victory, however, did not automatically secure political legitimacy. Yongle had to consolidate rule after what many contemporaries viewed as usurpation, while preserving the administrative capacity of a large examination-based bureaucracy.

This tension is central to our paper. A ruler in Yongle’s position did not need to choose between full continuity and complete purge. He could preserve the formal structure of meritocratic recruitment while using appointments and promotion to screen for political reliability. In the Ming case, obtaining the *jinshi* degree did not mechanically determine realized career attainment: subsequent appointments, postings, and access to the center remained administratively allocated and politically consequential. Bó and Chen (2026) document an institutional basis for this margin: during the First Ming procedure (1368–1594), entry-

level appointments were allocated through a sequential administrative assignment system in which candidates were ordered by degree and examination results, while posts were assigned by the Ministry of Personnel subject to eligibility and avoidance constraints. This institutional structure created a post-entry allocation margin through which political considerations could affect career trajectories. Comparative work on political transitions, cooptation, and bureaucratic organization highlights exactly such mixtures of continuity, selective incorporation, and political control (Acemoglu and Robinson, 2001; Gandhi and Przeworski, 2006; Jiang, 2018; Francois et al., 2015; Chen et al., 2025). Our empirical question is therefore not whether the regime punished all potential opponents uniformly, but where in the personnel pipeline political information was used most strongly.

2.2 From Loyalist Martyrs to County-Level Stigma

Our treatment is defined at the county level. We classify as treated those counties that produced officials historically recorded as Jianwen loyalist martyrs during or immediately after the Jingnan conflict. This definition is substantively meaningful. In late imperial China, local origin was a durable and legible identity, routinely recorded in administrative documents and widely recognized in elite politics. Native place was not merely a biographical detail: it was part of how officials were identified, evaluated, and connected within bureaucratic and elite life. After a civil war over dynastic legitimacy, the county that produced a martyr could serve as a shorthand marker for prior association with the defeated court, even for later *jinshi* who were themselves too young to have participated in the conflict.

This treatment therefore captures more than direct kin punishment. If the mechanism operated only through family ties, penalties should have remained concentrated within a narrow set of lineages. Our county-based definition instead captures the possibility that political memory attached to place and shaped later career allocation through collective beliefs about loyalty. Because direct lineage links are often unobservable in our data, a county-level marker is also the most historically grounded way to study how political suspicion may have extended beyond a small set of identifiable families. In that sense, the paper studies whether a regime transition generated a durable locality-based stigma within an otherwise continuous bureaucratic system.

This interpretation is also consistent with the language of early Yongle repression itself. In historical accounts of the persecution surrounding Jianwen loyalists, punishment was described not only in terms of core offenders and their families, but also as extending by association to the offenders' native localities. In the well-known case of Jing Qing, the record states that the authorities extended punishment to his native locality and allowed implication to spread outward, a process described as *gua man chao* ("plucking the melon vine"). For our purposes, the significance of this language is not that every martyr county was formally blacklisted in the same way, but that place of origin can become a politically legible marker through which suspicion extended beyond a narrow set of identifiable kin, as described in the *Mingshi jishi benmo* (Chronological Records of Important Events in Ming History).

2.3 Repression, Partial Rehabilitation, and the 1421 Capital Move

The historical record suggests that the early Yongle regime dealt harshly with core Jianwen loyalists and that punishment extended beyond the principal offenders themselves. Later sources indicate that by the Hongxi reign the court could explicitly describe the main offenders as already executed while ordering surviving family members who had been reduced to servile status to be restored to commoner status and have land returned. The language of clemency itself was limited: it concerned legal status and property rights, not a general restoration of official careers. This is important for our interpretation: it shows that early repression was real, but it also suggests that later clemency initially focused on family status and property rather than on a general restoration of bureaucratic prospects.

Subsequent softening remained partial. In 1457, during the Tianshun restoration, the court ordered the release and resettlement of Jianwen's descendants in Fengyang and provided stipends for them, but this was a controlled act of leniency toward the imperial line rather than a broad revision of appointment and promotion practices. Even this measure did not amount to political exoneration: the court's own language continued to preserve the earlier condemnation.

This gap between limited clemency and full political rehabilitation is historically plausible. As Ditmanson (2007) shows in his study of the mid- and late-Ming remembrance of the 1402 loyalists, discussion of these figures remained politically sensitive for a long period, and

more explicit commemoration emerged only gradually later in the Ming. That persistence matters for our argument because it implies that martyr origin was not merely a short-lived wartime label, but part of a politically charged field of memory that could continue to shape how local origins were interpreted within bureaucratic life. More explicit rehabilitation, including the restoration of the Jianwen reign title in official historiography, came well beyond our sample period ending in 1487. For this reason, symbolic rehabilitation need not imply that the personnel logic governing official careers had already changed.

The move of the capital from Nanjing to Beijing in 1421 provides an additional reason to distinguish symbolic memory from bureaucratic access. As Siu (2024) argues, the transfer of the capital had an explicitly political dimension: it reduced the influence of the civil government in the old capital and helped the Yongle emperor build a new political center of his own in the north. In that sense, the move was not merely geographic. It plausibly altered the structure of bureaucratic opportunity by increasing the value of trusted proximity to the court and weakening the position of officials more closely associated with the old political environment. We do not treat the capital move as mechanically identifying a single mechanism. Rather, it is best understood as a historically meaningful institutional shift that may have amplified the interaction between general southern disadvantage and county-specific political distrust. In the empirical analysis, this date marks a plausible moment at which locality-based political stigma could have become more consequential for later career advancement.

3 Empirical Strategy

3.1 Data and Sample

Our empirical analysis combines three types of information at the individual level: *jinshi* examination cohorts, county of origin, and subsequent career records. The unit of observation in the main specifications is an individual jinshi. For each jinshi, we observe the examination year, home county, examination ranking, and a set of realized career outcomes constructed from recorded official postings and promotions. The individual-level jinshi dataset is primar-

ily compiled from Qiu and Gong (2021), which rigorously collates and verifies a wide range of official records documenting career trajectories and additional biographical information. We further cross-checked the data against major examination registers and standard historical sources, including the *Mingshi* (History of Ming) and the *Ming Shilu* (Veritable Records of the Ming). The baseline sample spans examination cohorts from 1371 to 1487, covering both the pre-Jingnan and post-Jingnan periods within a broadly continuous institutional setting.²

The treatment is defined at the county level. An origin county is coded as treated if it produced at least one official historically recorded as a Jianwen loyalist martyr. Here, “martyr” refers to officials historically recorded as having died for the Jianwen regime during or immediately after the Jingnan conflict. The underlying logic is political trust: after a contested succession, counties associated with Jianwen loyalist martyrs could be seen as places tied to a defeated ruler whose supporters had directly challenged Yongle’s claim to the throne, making later officials from those counties less attractive for politically sensitive appointments and advancement.

Within the 599 counties represented in the *jinshi* birthplace sample, our baseline treatment group consists of 40 counties that produced at least one loyalist martyr who was executed or committed suicide rather than submitting to the Yongle regime. We identify these counties using martyrs’ records mainly from *Wuxue Bian: Jianwen Xunguo ChenJi* (Record of the Loyal Ministers of the Deposed Jianwen Regime).³

This treatment definition is historically meaningful but not plausibly random. It may also capture pre-existing county characteristics, such as local elite density, administrative importance, or stronger connections to the political center. Accordingly, the empirical design does not rely on random treatment assignment. Instead, identification comes from comparing changes in outcomes before and after 1402 between martyr and non-martyr counties,

²We end the sample in 1487 (the last *keju* examination held under the Chenghua Emperor) to preserve a sufficiently long window for observing post-entry careers while avoiding later cohorts whose careers would unfold under a less comparable political environment.

³This source was compiled before 1566 and is a biographical history of the early Ming Dynasty. We further cross-validate and supplement the information using other important near-contemporary sources written in the first half of the sixteenth century, including *Gechu Yishi* (Accounts of Events from the Reign of the Deposed Emperor), *Gechao Zhi* (Records of the Removed Regime), *Gechao Yizhong Lu* (Records of the Loyalists of the Removed Regime), *Beiyi Lu* (Record of Lost Matters), which record the individuals and events associated with the Jianwen reign and the Jingnan conflict.

while subjecting the estimates to a demanding set of pre-trend tests, geographic restrictions, matching exercises, and inference checks.

We study outcomes at two distinct stages of the bureaucratic pipeline. First, to examine whether the regime altered elite selection at entry, we construct an examination outcome equal to one if a *jìnshì* ranked in tier 1 or 2, which roughly corresponds to the top 30% of successful candidates in a given examination cohort. This measure captures whether candidates from martyr counties continued to perform well in the visible and formalized part of the elite recruitment system. At the same time, placement in the top two examination tiers normally translated into stronger initial appointments and therefore constituted an important early advantage in bureaucratic careers.

Second, to study post-entry career allocation, we focus on three main outcomes. The first is the highest official rank attained, coded on an 18-point scale. The second is an indicator for ending at a low rank, defined as rank 6 (Principal Seventh Rank) or below, which roughly corresponds to remaining at or below the county-magistrate level.⁴ The third is an indicator of ever holding a central government position. Together, these three measures capture complementary dimensions of career attainment: vertical promotion, confinement to lower offices, and access to the political center.

We restrict the baseline sample to *jìnshì* with birthplace information coded to standard county units. A subset of observations is recorded instead under *weisuo* (guard/battalion) jurisdictions rather than civil counties. Because *weisuo* units do not map consistently onto the county-level administrative geography used to define treatment, including them would introduce nontrivial ambiguity and additional measurement error in county assignment. We therefore exclude *weisuo*-coded birthplaces from the baseline county-level analysis.

Our baseline sample also excludes the 1385 cohort. By the time of the Jingnan conflict, the 1385 *jìnshì* had already been in office for many years and were directly exposed to the political struggle itself. They therefore capture the immediate personnel consequences of regime change for incumbent officials, rather than the post-entry career effects faced by later cohorts entering the bureaucracy after 1402. In addition, the surviving *jìnshì* information for

⁴Official ranks run from 9 to 1, with 1 denoting the highest status and 9 the lowest. Within each rank, principal (*zheng*) status is senior to vice/secondary (*cong*) status.

this cohort is materially less complete. Including the subset of individuals from this cohort for whom usable information is available attenuates the estimates but does not reverse their sign. We therefore exclude the 1385 cohort from the baseline analysis and report specifications including it as a robustness check.

The resulting *jinshi* sample contains 6,924 individuals, excluding observations recorded under *weisuo* jurisdictions and omitting cohort 1385. Among these, 6,347 can be matched to county-level birthplace information, and 5,665 have officeholding records. The baseline regression sample contains 5,448 observations; the difference is due to the exclusion of 217 singleton observations in specifications with fixed effects.

3.2 Difference-in-Differences Design

The baseline specification is a two-way fixed-effects difference-in-differences model (DiD):

$$Y_{ict} = \beta (Treat_c \times Post_t) + \alpha_c + \lambda_t + \varepsilon_{ict}, \quad (1)$$

where i indexes *jinshi*, c indexes county of origin, and t indexes examination year. $Treat_c$ equals one for martyr counties, and $Post_t$ equals one for examination cohorts after the Jingnan regime change. County fixed effects, α_c , absorb time-invariant differences across counties, while examination-year fixed effects, λ_t , absorb nationwide cohort shocks.

For examination outcomes, equation (1) tests whether *jinshi* from martyr counties experienced a differential shift in entry-stage performance after 1402. For career outcomes, the same specification estimates whether post-1402 cohorts from martyr counties faced differential changes in later advancement relative to comparable cohorts from other counties. Since the treatment is constant within county and the political rupture is national, the identifying variation comes from differential before-after changes between treated and untreated counties. The baseline regressions contain 599 county clusters, 40 of which are classified as martyr counties (Table 1). As shown in Figure 1, treated counties are concentrated in the South. We report standard errors clustered at the county level throughout the main analysis. Our identifying assumption is that, absent the post-1402 political shock, career outcomes in martyr and non-martyr counties would have evolved similarly, conditional on county and

cohort fixed effects and the included geographic interactions.

3.3 Dynamic Effects and the 1421 Break

The historical discussion in Section 2 suggests that the post-Jingnan penalty may not have been uniform over time. In particular, the move of the capital from Nanjing to Beijing in 1421 marks a plausible institutional break after which politically sensitive career allocation may have become more pronounced. We therefore estimate a period-split specification:

$$Y_{ict} = \beta_E (Treat_c \times Early_t) + \beta_L (Treat_c \times Late_t) + \alpha_c + \lambda_t + \varepsilon_{ict}, \quad (2)$$

where $Early_t$ denotes the post-1402 but pre-1421 period, and $Late_t$ denotes the 1421–1487 period. This specification tests whether the main post-1402 effect became stronger after 1421, when the political center shifted from Nanjing to Beijing.

To complement the split-period estimates, we also estimate event-study specifications:

$$Y_{ict} = \sum_{\tau \neq \tau_0} \beta_\tau (Treat_c \times 1[t = \tau]) + \alpha_c + \lambda_t + \varepsilon_{ict}, \quad (3)$$

where the omitted period τ_0 is the reference period. In the binned specification, we use 1394–1400 as the omitted reference period and estimate one pre-period bin (1371–1391) and seven post-period bins. The bins are constructed to balance dynamic resolution against precision: cohorts are grouped within historically meaningful Ming sub-periods, with cuts at the 1402 regime change and the 1421 capital relocation, and post-1421 bins follow Ming reign boundaries.⁵

3.4 Geographic Controls and Comparative Samples

A central concern is that martyr counties may differ systematically from other counties along dimensions that also shape later careers. We address this concern in several ways. First,

⁵The post-1421 reign-based bins correspond to 1439–1454 (Zhengtong–Jingtai), 1457–1464 (Tianshun), and 1466–1475 / 1478–1487 (early and late Chenghua). Two bins deserve clarification: 1371–1391 is grouped because individual cohorts contain only 15–27 treated-county *jinshi*, and 1421–1436 combines the 1421–1427 and 1430–1436 sub-windows to maintain precision in the immediate post-relocation period.

we control for time-invariant geographic characteristics interacted with the post indicator, including an indicator for southern location and log distances to Nanjing and Beijing. This specification allows broad geographic gradients to evolve differentially after 1402, rather than being absorbed by the treatment effect. In an additional robustness check, we further control for log suitability for major crops (rice and wheat) and log county population. The only county-level population data available come from the *Da Ming Yi Tong Zhi* (Comprehensive Gazetteer of the Great Ming), compiled in 1461, near the end of our sample period.

Second, as a robustness check, we restrict the sample to southern provinces, where most martyr counties were located and where the relevant political geography was highly asymmetric. Third, we construct matched samples based on pre-period *jinshi* intensity. Specifically, we restrict the control group to counties whose pre-1400 *jinshi* counts fall within a specified bandwidth of those in treated counties. Although this does not yield a fully experimental comparison, it helps address the concern that martyr counties may simply have been more elite or politically salient at baseline. Finally, we report specifications with prefecture fixed effects and prefecture-level clustering. This absorbs broader prefectural heterogeneity and allows us to retain observations for individuals for whom county information is missing while prefecture is known.

4 Empirical Results

4.1 From Examination Entry to Career Allocation

This paper contrasts outcomes at two stages of the bureaucratic pipeline: formal elite entry through the *jinshi* examination, and subsequent career allocation within the bureaucracy. In the Ming bureaucratic system, obtaining the *jinshi* degree marked formal entry into the elite recruitment pipeline, while subsequent appointments, postings, and promotions determined realized career attainment. If the post-Jingnan regime broadly excluded candidates from martyr counties, we should observe deterioration already at the examination stage. If, instead, the regime preserved visible meritocratic recruitment while screening more heavily in appointments and promotion, the main penalties should appear mostly after entry, in the

career outcomes.

Table 1 already suggests why this distinction matters. Before 1402, martyr counties were not a visibly disadvantaged group. In the pre-period, their *jinshi* were more likely to place in the top two examination tiers (0.411 vs. 0.283), less likely to end at low rank (0.424 vs. 0.596), and more likely to hold central positions than *jinshi* from other counties (0.685 vs. 0.545), while the difference in highest lifetime rank is modest and statistically weak (8.20 vs. 7.49). Treated counties were also more concentrated in the South, and they were larger on average in the available county population data, which are measured around 1461 rather than in the pre-Jingnan period. These baseline differences matter substantively because they imply that the post-1402 estimates do not reflect a mechanically disadvantaged group continuing to perform poorly. If anything, martyr counties appear relatively strong before the political rupture. Table 1 therefore makes the later reversal more notable.

We first examine whether the rupture altered entry-stage performance. Table 2 shows little evidence that *jinshi* from martyr counties suffered a broad post-1402 decline in examination rankings. In the baseline specification, the coefficient on $Treat_c \times Post_t$ is negative but small and statistically insignificant. The same is true when the sample is restricted to southern counties and when geographic interactions are added. In the dynamic specification, the early post-1402 coefficient is positive and imprecisely estimated, while the late coefficient is modestly negative and statistically insignificant (-0.049). Since placement in the top two tiers often translated into a higher starting point in official appointment, the absence of an entry-stage penalty provides a useful counterfactual benchmark: absent additional political screening after entry, post-1402 career outcomes for martyr-county *jinshi* should have remained broadly similar to their pre-Jingnan baseline.

Panel A of Figure 2 provides the corresponding dynamic evidence for examination placement. The estimates show no post-1402 collapse in the probability of placement in examination tiers 1–2. If anything, the coefficient for the 1404–1412 cohorts is positive and statistically distinguishable from zero, indicating that candidates from martyr counties continued to succeed at the examination stage in the years immediately following the regime change. Although the pooled Early coefficient in Table 2 is statistically imprecise, the binned estimates show that this positive differential is concentrated in the 1404–1412 cohorts and

fades thereafter, with later post-1402 coefficients statistically indistinguishable from zero. Taken together, these results suggest that the new regime did not substantially alter who could succeed in the publicly observed examination stage. This pattern contrasts with the career outcomes examined below: it points away from overt exclusion at entry and toward screening through later bureaucratic allocation.

Table 3 presents the baseline estimates for post-entry career outcomes. The estimates imply a decline of about 1.039 rank points on the 18-point rank scale, a 17.7 percentage point increase in the probability of ending at rank 6 or below, and a 12.6 percentage point decline in the probability of ever serving in the center. The central-access result should be interpreted more cautiously than the two rank-based outcomes: the year-by-year pre-trend test rejects the null of parallel pre-trends for ever holding a central position at the 5 percent level ($p = 0.046$), whereas the corresponding tests for highest official rank and low-rank attainment do not. We therefore place greater weight on the rank-based outcomes, while treating central access as directionally consistent supporting evidence.

4.2 Dynamic Effects and the Post-1421 Intensification

The historical discussion in Section 2 suggests that the post-Jingnan penalty may have strengthened after 1421, when the capital moved from Nanjing to Beijing. Equation (2) formalizes this possibility by allowing the post period to differ between 1404–1418 and 1421–1487, while equation (3) provides a more flexible event-study view.

Table 4 shows that the early post-1402 period was relatively muted. For highest official rank, the early coefficient is negative but imprecisely estimated, while the late coefficient is much larger in magnitude and statistically significant, at -0.354 and -1.305, respectively. A similar pattern appears for low-rank attainment. The early coefficient is positive but imprecisely estimated, whereas the late coefficient is larger and statistically significant, at 0.100 and 0.207, respectively. For central access, the contrast between the two periods is less pronounced. In the baseline split, the early coefficient is negative and marginally significant (-0.116), but it becomes smaller once southern period effects are added (-0.065). The late coefficient remains negative and statistically significant in the last two columns (-0.130 and -0.077). Overall, these estimates suggest that the main deterioration in career attainment

was concentrated after the move to Beijing rather than immediately after the 1402 political rupture. This pattern is especially clear for rank outcomes, while for central access, part of the decline appears to reflect a broader south-wide disadvantage after the capital shift.

Panels B–D of Figure 2 provide the corresponding event-study evidence for career outcomes. Although the post-1421 patterns in the binned event studies may visually resemble Panel A, the pooled estimates in Tables 2 and 4 reveal a clearer distinction: the late-period coefficient on Tier 1–2 placement is small and statistically insignificant (-0.049), whereas the late-period coefficients for the rank-based career outcomes are larger in magnitude and statistically significant. The event-study panels are individually noisier than the pooled estimates, but they document the timing of this divergence. In Panel B, the coefficients for highest official rank are close to zero in the pre-1421 post bins, turn negative after the capital relocation, and remain negative through the end of the sample. The individual post-1421 bin estimates are less precise than the pooled late-period estimate in Table 4, but their signs are uniformly consistent with the split-period result.

In Panel C, the coefficients for low-rank attainment turn positive after 1421 and remain positive across all subsequent bins, with several individually distinguishable from zero. This provides the clearest graphical counterpart to the split-period estimate of $+0.207$ in Table 4. Panel D shows a different timing pattern for central access. The coefficients become negative already in the 1415–1418 cohorts, before the capital relocation, and remain negative thereafter. This earlier onset is consistent with the table-based evidence that central access is less tightly tied to a clean post-1421 break and may partly reflect broader southern disadvantage after the capital shift.

The dynamic pattern also alleviates a related concern about the direct effects of war itself. One possibility is that early Yongle *jinshi*, especially those from politically sensitive counties, were disproportionately assigned to war-damaged or otherwise less desirable posts in the immediate aftermath of the Jingnan conflict, which could mechanically depress later career outcomes. The estimates do not support this as the main explanation. If anything, the overall penalty is smaller and less precisely estimated among the earliest post-1402 cohorts, while the stronger and more consistent effects emerge later.⁶

⁶Moreover, it is possible that some *jinshi* may have originated from counties directly affected by the

Overall, the dynamic evidence suggests that the main career penalty became more pronounced after 1421 rather than immediately after 1402. By itself, this pattern does not identify a single mechanism. It does, however, point away from a one-time postwar purge and toward a more persistent penalty operating within the bureaucratic career ladder.

The dynamic pattern also raises a natural question: Does the stronger post-1421 penalty simply reflect the move of the capital to Beijing and a broader deterioration in southern officials' prospects, or did martyr counties face an additional disadvantage beyond this general shift?

4.3 County-Level Stigma Beyond Capital Relocation

Table 5 shows that the stronger post-1421 penalty cannot be reduced to a general deterioration in southern officials' prospects after the capital moved to Beijing. Panel A compares southern martyr counties with other southern counties, using northern counties as the omitted category. In the baseline static specification, the contrast is sharp for career-rank outcomes. Southern non-martyr counties do not exhibit comparable post-1402 deterioration in the highest official rank or low-rank attainment (-0.353 and 0.054, respectively), whereas southern martyr counties do (-1.314 and 0.219), and the equality tests reject equality across the two groups. This pattern is difficult to reconcile with a purely geographic account in which all southern counties were affected similarly by the capital shift.

The dynamic specification further weakens a purely geographic interpretation. If the move to Beijing simply imposed a common disadvantage on southern officials, southern martyr and non-martyr counties should evolve more similarly over time, with the main contrast operating between the South and the North. Instead, the sharpest divergence appears within the South and is concentrated in the late period. For the highest official rank and low-rank attainment, *Southern martyr* $\times$ *Late* is -1.563 and 0.247, compared with

Jingnan civil war. If wartime disruption damaged local education, elite networks, or public administration, then part of the estimated post-1402 career penalty could reflect exposure to war rather than the political stigma attached to martyr-county origin. To address this possibility, we exclude officials from counties located in areas directly involved in the conflict; nearby areas are also treated as war-affected when geographic proximity makes spillovers likely. The resulting sample still contains 5,430 observations, and the estimated effects on the three career outcomes remain very robust. This makes it less likely that the main findings are driven simply by wartime destruction or local disruption in conflict-exposed areas.

-0.315 and 0.050 for *Southern non-martyr* $\times$ *Late*, and the equality tests reject equality in the late period. By contrast, the early-period differences are smaller and not statistically distinguishable. The key implication is therefore not just that the penalty intensified over time, but that the widening gap followed martyr-county lines within the South, which capital relocation alone does not explain.

The evidence is more mixed for central access. In the baseline specification, both southern martyr and southern non-martyr counties experienced post-period declines in ever holding a central position relative to the north, and the difference between the two groups is smaller and only marginally significant (-0.310 versus -0.236). The dynamic estimates point in the same direction: both groups exhibit negative coefficients in the early and late periods, while the equality tests do not reject equality across martyr and non-martyr counties. This is consistent with Table 4, which suggests that broader southern disadvantage appears more relevant for central appointments than for rank outcomes. Restricting attention to southern counties only, martyr counties still diverge from other southern counties after 1402, especially in the highest rank and low-rank attainment, consistent with the within-South divergence documented in Panel A. In sum, Panel A indicates that capital relocation may help explain a broader southern disadvantage in central access, but it cannot account for the sharper deterioration in rank outcomes for martyr counties.

Panel B provides a complementary test based on distance to Beijing. In the baseline specification, the triple interaction between $Treat_c \times Post_t$ and log distance to Beijing is positive and significant for highest official rank (8.352) and negative and significant for low-rank attainment (-0.581), indicating that the martyr-county career penalty attenuated with distance from the new capital. Since nearly all treated counties are located in the South, this gradient is identified primarily from variation in distance among southern counties rather than from a comparison between northern and southern treated counties. The estimates are therefore best interpreted as showing that the relative post-1402 penalty for martyr counties was stronger for counties relatively closer to Beijing and weaker for counties farther away, rather than as precise estimates of the effect at any particular distance.

Overall, capital relocation alone cannot account for the observed pattern. Panel A shows that martyr counties performed substantially worse than other southern counties, especially

in the later post period, which a uniform southern disadvantage does not explain. Panel B suggests that the penalty attenuated with distance from Beijing, though this gradient is identified from limited spatial variation. The combined evidence points to county-level political stigma that became more consequential after the political center shifted to Beijing.

4.4 County Stigma Versus Narrow Family Retribution

Another important question is whether the post-1402 career penalty reflected the broad political stigma attached to martyr counties or a narrower form of family retribution directed at the descendants and close kin of Jianwen loyalists. These two interpretations have different implications. If the penalty operated mainly through lineage-based punishment, later *jinshi* from the martyr counties and shared surnames with recorded martyrs should have faced substantially worse career outcomes than other *jinshi*. If, instead, political suspicion attached more broadly to the county of origin, then the penalty should extend beyond the most directly connected families.

To distinguish between these possibilities, we divide *jinshi* from martyr counties into two groups: those who shared a surname with at least one recorded martyr and those who did not. We then estimate the baseline and dynamic specifications separately for the two groups using the same specification. Although surname is an imperfect proxy for kinship, this comparison is informative about whether the post-1402 penalty was concentrated within a narrow set of lineages or extended more broadly to officials from politically marked localities.

The results in Table 6 support the broader county-stigma interpretation. In the static specification, both same-surname and different-surname officials from martyr counties experienced substantial post-1402 penalties. For the highest official rank, the coefficients are -0.874 and -1.393, respectively. For low-rank attainment, the corresponding effects are 0.159 and 0.215. For central access, both groups again experienced similar declines, at -0.136 and -0.104. Overall, the equality tests do not support the view that the same-surname group was systematically more heavily penalized. For the highest official rank, if anything, the coefficient is larger in magnitude for the different-surname group, although the difference is only marginally significant ($p = 0.087$).

The dynamic decomposition points in the same direction. For both the same-surname

and different-surname groups, the early-period coefficients are comparatively small and imprecisely estimated. For the same-surname group, the late-period coefficients are -1.139 for the highest official rank, 0.190 for low-rank attainment, and -0.143 for central access. For the different-surname group, the corresponding estimates are -1.629, 0.239, and -0.105. Thus, the stronger post-1421 deterioration was not confined to officials who shared surnames with recorded martyrs.

These results are more consistent with a broader county-level stigma than with a narrow family-retribution account. The relevant political marker appears to have extended beyond the family itself. Once a county became associated with Jianwen loyalist martyrdom, later *jinshi* from that locality faced worse career outcomes even when they did not share the martyrs' surnames. Together with Table 5, this pattern suggests a durable, place-based political stigma rather than either a purely geographic effect or a narrowly targeted lineage sanction.

4.5 Pre-1402 *jinshi* Output and the Martyr-County Penalty

A natural concern is that the post-1402 penalty may reflect pre-existing differences in local bureaucratic capacity rather than place-based political stigma. If martyr counties had weaker examination pipelines before the Jingnan rupture, their later career disadvantage might partly capture differences in local elite reproduction rather than differential treatment within the bureaucracy. To address this concern, we ask whether the post-1402 penalty varies with a county's pre-existing elite pipeline. Specifically, we interact $Treat_c \times Post_t$ with the log number of pre-1402 *jinshi* produced by county c , while also including $Post_t \times \log(1 + PreJinshi_c)$ to absorb general post-period changes associated with counties that had stronger pre-1402 *jinshi* output.⁷

Table 7 reports the results. Two patterns are worth emphasizing. First, the penalty does not vanish among counties with substantial pre-1402 elite pipelines. Evaluated at the

⁷We also estimate specifications using raw county-level *jinshi* counts, prefecture-level pre-1402 *jinshi* output, and a dynamic split between the early (1404–1418) and late (1421–1487) post periods. These additional specifications yield qualitatively similar conclusions and are reported in Table A1. An additional heterogeneity check using recorded county population, which is observed around 1461 rather than before the Jingnan rupture, is reported in Appendix A.1.

treated-county mean of pre-1402 *jinsshi* output, the penalty remains -1.5 ranks for highest official rank ($p < 0.01$), +18 percentage points for low-rank attainment ($p < 0.01$), and -14 percentage points for ever holding a central position ($p < 0.01$). This makes it unlikely that the result is driven solely by martyr counties having weaker prior elite pipelines: counties with substantial prior pipelines were also penalized.

Second, the penalty’s magnitude on highest official rank is significantly moderated by prior *jinsshi* output. The triple interaction is positive and significant for highest official rank ($p = 0.002$); for low-rank attainment, the interaction is negative but imprecise ($p = 0.13$); and for central appointment, the interaction is not statistically distinguishable from zero ($p = 0.36$). We do not detect a statistically significant difference in this moderating effect between the early (1404–1418) and late (1421–1487) post-1402 periods. We interpret pre-1402 *jinsshi* output as a reduced-form proxy for several correlated county characteristics: educational capacity, bureaucratic embeddedness, and local elite density, rather than as a direct measure of any specific channel such as patronage. Within this caveat, the central implication is that the martyr-county penalty operated on top of, rather than because of, differences in pre-existing elite pipelines: prior *jinsshi* output did not eliminate the penalty, but it was associated with a smaller disadvantage on rank trajectories.

4.6 Additional Robustness Evidence

We conduct a broad set of robustness checks to assess whether the main findings are sensitive to inference procedures, placebo timing, sample construction, or the choice of comparison group. Table 8 summarizes the main results.

Panel A provides several inference and placebo diagnostics. Wild-cluster bootstrap p -values are 0.022, 0.004, and 0.007 for the highest official rank, low-rank attainment, and central access, respectively. County-level randomization inference leads to a similar conclusion. Specifically, we repeatedly reassign treatment status by replacing treated counties with control counties and compare the actual estimate with the resulting placebo distribution. The corresponding randomization-based p -values are 0.067, 0.014, and 0.022, again indicating that the main findings are unlikely to be artifacts of the small number of treated counties. We also assign a fake post period within the pre-Jingnan era, treating 1391–1400 as

a placebo post period, and re-estimate the baseline specification. The placebo $Treat \times Post$ coefficients are small and statistically insignificant for all three outcomes, which helps rule out the concern that the results merely reflect pre-existing divergence unrelated to the regime change. Panel A also reports pre-trend diagnostics. For highest official rank and low-rank attainment, both the binned pre-trend tests and the year-by-year pre-trend F-tests are insignificant. For central access, the year-by-year pre-trend test is noisier.

Panel B shows that the results are not driven by a particular sample definition. Including the 1385 cohort attenuates the baseline coefficients, but the overall pattern remains similar. The attenuation is unsurprising because surviving career information for the 1385 cohort is materially less complete. In unreported split-period estimates for this expanded sample, the late-period coefficients remain robust for all three career outcomes, at -0.996 for highest official rank, 0.117 for low-rank attainment, and -0.110 for central access (p -values are 0.024, 0.033, and 0.011, respectively), whereas the early-period coefficients remain small and imprecise.

Restricting the sample to the South also preserves the qualitative pattern. The coefficients for highest official rank and low-rank attainment remain quantitatively similar to the baseline estimates, although the effect on central access becomes notably smaller, consistent with the earlier discussion that part of the baseline central exclusion may reflect broader North–South differences in political geography.

The matched-sample results further strengthen the baseline interpretation. Here we restrict the sample to counties with relatively similar pre-period *jinshi* output, so that treated and control counties are more comparable in underlying educational capacity. Across matching cutoffs of 5, 10, and 20, the coefficients remain negative for highest official rank and central access and positive for low-rank attainment. If anything, the estimates often become larger in magnitude as the comparison set is tightened. The low rank attainment results are also not sensitive to modest changes in outcome definition or additional controls. Using alternative low-rank cutoffs (≤ 7 or ≤ 8) yields the same qualitative conclusion. Adding geographic controls interacted with Post, including agriculture and population, leaves the main pattern unchanged. The estimates are also robust to controlling for examination tier 1–2 placement, a proxy for individual ability that itself shows no systematic post-1402 shift in

martyr counties (Table 2), suggesting that the career penalty is unlikely to reflect underlying ability differences.

A related concern is that the baseline results might reflect broader regional heterogeneity rather than treatment itself. Table A2 addresses this by replacing county fixed effects with prefecture fixed effects and clustering at the prefecture level, thereby restricting identification to within-prefecture comparisons while also retaining officials whose county is unknown but prefecture is observed. Even under this tighter geographic comparison, post-1402 *jinshi* from martyr counties continued to attain lower highest official rank, were more likely to remain at low rank, and were less likely to hold central positions (-0.836, 0.148, and -0.139).

4.7 Record Completeness

An important concern in historical career data is record completeness. Because the data are compiled from official histories and related historical sources, they should not be interpreted as mechanically complete for every individual or every stage of a bureaucratic career. The relevant question is therefore not whether the surviving record is exhaustive, but whether the main results can plausibly be explained by missing records that are systematically correlated with martyr-county origin after 1402.

Several features of the evidence make that interpretation less likely. First, the paper's main outcomes concern post-entry careers rather than entry into the examination system. If the core findings were driven primarily by selective disappearance from the historical record, one might expect this to appear first in whether officials are observed at all. Yet the selection checks do not indicate differential post-1402 effects on whether a *jinshi* has any recorded office (-0.014, p -value = 0.773, and -0.043, p -value = 0.271 even when the 1385 cohort is included). This does not eliminate all concerns about record incompleteness, but it suggests that the main findings are not simply the consequence of martyr-county officials dropping out of the observable record altogether.

Second, the main rank outcome is the highest observed office attained. Even if historical records omit minor postings or intermediate career steps, later and more prestigious appointments are generally more salient and therefore more likely to be preserved than routine administrative details. Conditional on being observed in office at all, it is thus less plausible

that the paper’s central result on the highest official rank is driven entirely by the selective omission of subsequent, more prominent appointments.

More broadly, a missing-record explanation would require a fairly specific pattern of selective survival: after 1402, officials from martyr counties would need to remain similarly likely to appear in the record at least once, and similarly able to succeed at the examination stage, yet be systematically less likely to have their subsequent career advancement preserved in surviving sources. Such a mechanism cannot be ruled out completely, but it is narrower and less naturally implied by general concerns about record incompleteness. For that reason, record incompleteness is unlikely to provide a full explanation for the post-entry career penalties documented here.

A related concern is that record incompleteness may operate not only through missing career observations, but also through incomplete identification of treated units, that is, through missing martyrs from counties not identified as treated. Given the historical context, it is possible that the baseline treatment group of 40 martyr counties is itself undercounted, so that some truly treated counties are mistakenly classified as controls. This corresponds to false negatives in the treatment indicator and, under standard assumptions, would bias the estimates toward zero.

To assess this possibility, we conduct a Monte Carlo exercise that imposes a deliberately pessimistic misclassification scenario. In each of 1,000 iterations, we randomly draw a number K from a uniform distribution $U[1, 30]$, representing the unknown number of genuinely treated counties that were not identified in the baseline data. We then randomly select K counties from the untreated group, temporarily reassign them to treatment, and re-estimate the baseline specifications under this alternative treatment coding.

The results indicate that the main findings are robust to this form of county-level undercounting. Even when treatment status is diluted by randomly reclassifying as many as 30 counties from the untreated group, the estimated effects remain statistically significant at the 10% level in the overwhelming majority of simulations: 98.8% for *high-rank*, 98.3% for *ever-central*, and 99.9% for *low-rank*. Likewise, the 90% ranges of the simulated coefficients continue to exclude zero for all three outcomes ($[-1.149, -0.780]$ for *high-rank*, $[-0.140, -0.088]$ for *ever-central*, and $[0.133, 0.190]$ for *low-rank*).

Figure A.1 complements these summary statistics by showing how the estimated coefficients evolve as the number of reassigned counties increases. For each outcome, the figure plots the mean simulated coefficient (solid line) together with the 90% confidence band (shaded area) against the number of counties added to the treatment group. As expected under attenuation bias, the mean estimates drift modestly toward zero as more potentially misclassified counties are added to the treatment group. Even so, the simulated coefficients remain close to the original baseline estimates (red dashed lines), and the confidence bands do not cross zero for any of the three outcomes.

Collectively, these results suggest that record incompleteness is unlikely to overturn the paper’s substantive conclusions. The individual-level evidence indicates that the main findings are not driven simply by martyr-county officials disappearing from the observable record, while the county-level simulation shows that even substantial under-identification of treated counties would mainly attenuate the estimates rather than eliminate them. Both forms of evidence therefore point in the same direction: incomplete historical records may introduce noise, but they are unlikely to account for the post-entry career penalties.

5 Conclusion

This paper shows that the political costs of loyalty to a defeated ruler need not operate through exclusion from elite entry. In the aftermath of the Jingnan Rebellion, later *jinshi* from counties associated with Jianwen loyalist martyrs were not systematically screened out at the point of examination success. Instead, the penalty emerged after entry, through slower advancement, lower peak rank, and weaker access to the upper reaches of the bureaucracy. We refer to this pattern as displaced political screening: the formal structure of meritocratic recruitment was preserved, while political trust was reallocated at later, less visible stages of career allocation.

The disadvantage became more pronounced after 1421, when the capital moved to Beijing and proximity to the political center became more consequential, especially for rank-based outcomes. The evidence is also more consistent with locality-based stigma than with narrow lineage-based retaliation: later *jinshi* from martyr counties faced similar disadvantages re-

gardless of whether they shared a surname with a recorded loyalist martyr. What persisted, then, was not simply punishment directed at identifiable families, but a durable political reputation attached to place.

More broadly, these findings speak to a problem that extends beyond imperial China. States that retain examinations, credentialed recruitment, or other formalized channels of elite selection often face a recurring tension between administrative competence and political trust. One way to manage that tension is not to bar entry openly, but to shift screening later and less visibly into the allocation of careers: who is promoted, who remains peripheral, and who gains access to the political center. The Ming case thus highlights a mode of regime consolidation that preserves the formal appearance of meritocratic recruitment while reallocating trust within inherited institutions.

For studies of bureaucratic selection more broadly, our analysis suggests that formal access and realized advancement should be analyzed as distinct margins of political control. This perspective connects the Ming evidence to broader research on bureaucratic allocation, political loyalty, and the competence–trust trade-off in personnel systems. The relevant margin is not only who enters the state, but how far different groups are allowed to advance once inside it.

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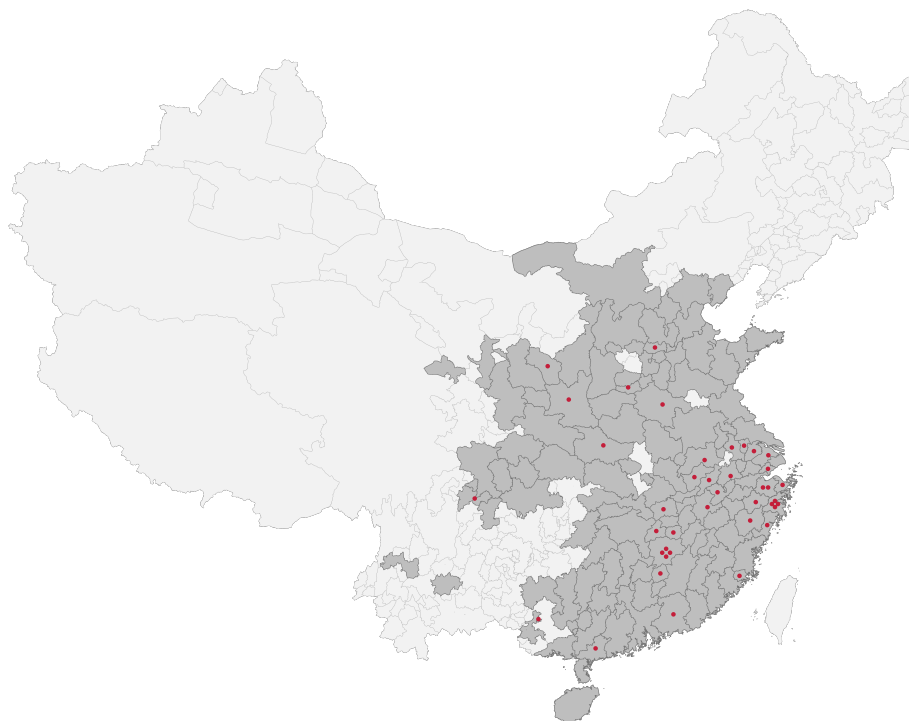
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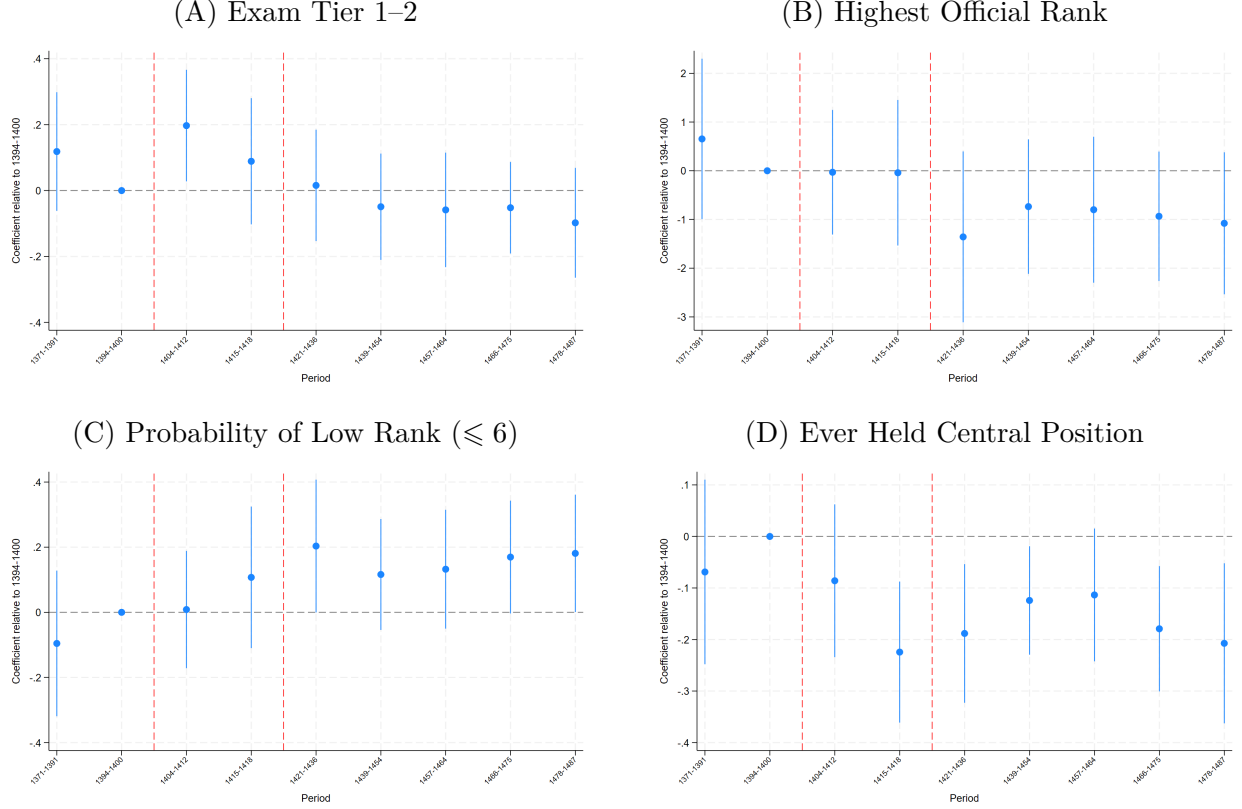
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Figure 1: Illustrative Distribution of Treated Counties in the Baseline Sample



Note: The base map is drawn from the Hartwell China Historical GIS 1391 prefecture-level dataset, which approximates Ming prefectural boundaries circa 1391. Red dots denote the number of treated counties in each prefecture in the baseline regression. Shaded areas indicate prefectures with at least one *jinshi*-producing county in the baseline regression. Because the map is drawn at the prefectural level, not all counties in shaded prefectures produced *jinshi*. Our sample consists of only 599 counties that produced *jinshi*, including 40 treated counties.

Figure 2: Binned Event Studies across Examination and Career Outcomes



Note: Each panel reports coefficients from a binned event-study specification with 95% confidence intervals, relative to the omitted period 1394–1400. Adjacent examination cohorts are grouped into bins defined by the political and administrative breaks of the early-to-mid Ming: 1371–1391 (early pre-period, combined to reduce noise from sparse cohorts), 1394–1400 (the three cohorts immediately preceding the Jingnan conflict, used as the omitted baseline), 1404–1412 (post-1402 cohorts under the capital at Nanjing), 1415–1418 (later Yongle cohorts, still pre-relocation), 1421–1436 (the first cohorts after the capital moved to Beijing), 1439–1454 (Zhengtong–Jingtai), 1457–1464 (Tianshun), 1466–1475 (early-to-mid Chenghua), and 1478–1487 (late Chenghua). Red dashed vertical lines mark 1402 and 1421. Standard errors are clustered at the county level.

Table 1: Descriptive Statistics

	Full Sample	Martyr Counties (Treat=1)	Other Counties (Treat=0)
Panel A: Sample Structure			
Total <i>jinshi</i>	5,448	943	4,310
Number of counties	599	40	559
Number of exam cohorts	35	35	35
Pre-period <i>jinshi</i> (1371–1400)	424	92	332
Post-period <i>jinshi</i> (1404–1487)	5,024	851	4,173
Panel B: Pre-Period Outcome Means			
Exam tier 1–2 share	0.310	0.411	0.283**
Highest rank (1–18)	7.65	8.20	7.49*
Low rank (≤ 6) share	0.559	0.424	0.596***
Ever central position share	0.575	0.685	0.545**
Panel C: County Characteristics			
Share southern	0.636	0.846	0.613**
Log distance to Nanjing	13.10	12.94	13.12
Log distance to Beijing	13.67	13.94	13.64**
Population (Tianshun era)	105.7	227.5	92.1***

Notes: Panel A reports the number of jinshi, counties, and exam cohorts. The 1385 cohort is excluded throughout. Panel B reports mean outcomes in the pre-period (exam years 1371–1400). Panel C reports county-level characteristics, measured at the county-first-appearance level. Asterisks indicate significance of two-sample t-tests for equality of means between treat and control: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Effect on Examination Rankings (Outcome: Tier 1–2 Share)

	(1) Baseline	(2) South Only	(3) + Geo Controls	(4) Early/Late
Treat $\times$ Post	-0.067 (0.065)	-0.065 (0.065)	-0.100 (0.068)	
Treat $\times$ Early (1404–1418)				0.065 (0.056)
Treat $\times$ Late (1421–1487)				-0.049 (0.035)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	5,448	5,253	4,339	5,448
Clusters	599	583	347	599

Notes: The dependent variable is an indicator equal to one if the *jinshi* ranked in tier 1 or 2. The 1385 cohort is excluded throughout. Column (2) restricts the sample to southern provinces. Column (3) adds South $\times$ Post, log(distance to Nanjing) $\times$ Post, and log(distance to Beijing) $\times$ Post as controls. Column (4) splits the post period into Early (1404–1418) and Late (1421–1487, after the capital move to Beijing) to test whether the absence of an entry penalty holds in both sub-periods. Standard errors clustered at the county level are reported in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 3: Career Outcomes of *jinshi* from Martyr Counties

	(1) Baseline	(2) + Geo × Post	(3) South Only	(4) Matched Sample
Panel A: Highest Official Rank (1–18 scale)				
Treat × Post	-1.039** (0.435)	-0.992** (0.428)	-0.937** (0.445)	-1.935*** (0.644)
Observations	5,448	5,448	4,339	3,174
Clusters	599	599	347	305
Control mean (pre)	7.49	7.49	7.49	7.53
Panel B: Low Rank (≤6, County Magistrate Level)				
Treat × Post	0.177*** (0.052)	0.172*** (0.053)	0.159*** (0.054)	0.237*** (0.073)
Observations	5,448	5,448	4,339	3,184
Clusters	599	599	347	305
Control mean (pre)	0.60	0.60	0.58	0.60
Panel C: Ever Held Central Position				
Treat × Post	-0.126*** (0.045)	-0.077* (0.044)	-0.070* (0.042)	-0.151** (0.069)
Observations	5,448	5,448	4,339	3,184
Clusters	599	599	347	305
Control mean (pre)	0.55	0.55	0.59	0.57
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: The 1385 cohort is excluded throughout. Panel A uses the official-rank outcome and the matched-sample specification in column (4). Column (2) adds South × Post, log(distance to Nanjing) × Post, and log(distance to Beijing) × Post. Column (3) restricts the sample to southern provinces. Column (4) matches counties on pre-period *jinshi* count (1–10). Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Early (Nanjing) vs. Late (Beijing) Period Effects

Outcome:	(1) Highest Official Rank	(2) Highest Official Rank	(3) Low Rank (≤ 6)	(4) Low Rank (≤ 6)	(5) Ever Held Central Position	(6) Ever Held Central Position
Treat $\times$ Early (1404–1418)	-0.354 (0.457)	-0.290 (0.463)	0.100 (0.064)	0.090 (0.065)	-0.116* (0.062)	-0.065 (0.061)
Treat $\times$ Late (1421–1487)	-1.305*** (0.471)	-1.231** (0.482)	0.207*** (0.054)	0.195*** (0.056)	-0.130*** (0.044)	-0.077* (0.043)
South $\times$ Early (1404–1418)		-0.227 (0.641)		0.043 (0.094)		-0.248*** (0.084)
South $\times$ Late (1421–1487)		-0.334 (0.496)		0.052 (0.071)		-0.233*** (0.072)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,448	5,448	5,448	5,448	5,448	5,448
Clusters	599	599	599	599	599	599

Notes: Early = 1404–1418 (capital in Nanjing); Late = 1421–1487 (capital in Beijing). Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Capital Relocation, Political Stigma, and Distance to the New Capital

Outcome:	Baseline Static Specification			Dynamic Split Specification		
	(1) Highest Official Rank	(2) Low Rank (≤ 6)	(3) Ever Held Central Position	(4) Highest Official Rank	(5) Low Rank (≤ 6)	(6) Ever Held Central Position
Panel A: Southern Counties Comparison						
Southern non-martyr $\times$ Post	-0.353 (0.499)	0.054 (0.072)	-0.236*** (0.072)			
Southern martyr $\times$ Post	-1.314** (0.595)	0.219*** (0.078)	-0.310*** (0.076)			
Southern non-martyr $\times$ Early				-0.327 (0.642)	0.054 (0.095)	-0.250*** (0.085)
Southern non-martyr $\times$ Late				-0.315 (0.497)	0.050 (0.071)	-0.232*** (0.072)
Southern martyr $\times$ Early				-0.574 (0.729)	0.139 (0.105)	-0.309*** (0.096)
Southern martyr $\times$ Late				-1.563** (0.618)	0.247*** (0.078)	-0.312*** (0.076)
Equality test (p -value)	0.031	0.002	0.094			
Equality test, Early (p -value)				0.595	0.201	0.336
Equality test, Late (p -value)				0.010	0.001	0.065
Panel B: Distance to Beijing						
Treat $\times$ Post	-4.096*** (1.372)	0.384*** (0.119)	-0.182* (0.098)			
Treat $\times$ Post $\times$ log distance to Beijing	8.352** (3.375)	-0.581** (0.272)	0.229 (0.247)			
Treat $\times$ Early				-3.288** (1.422)	0.270* (0.147)	-0.208* (0.108)
Treat $\times$ Late				-4.365*** (1.422)	0.420*** (0.122)	-0.176* (0.098)
Treat $\times$ Early $\times$ log distance to Beijing				8.007** (3.460)	-0.465 (0.327)	0.336 (0.256)
Treat $\times$ Late $\times$ log distance to Beijing				8.337** (3.529)	-0.600** (0.288)	0.198 (0.255)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,448	5,448	5,448	5,448	5,448	5,448
Clusters	599	599	599	599	599	599

Notes: The omitted category in Panel A is the northern counties. Columns (1)–(3) report the baseline post-1402 specification, while columns (4)–(6) report the dynamic specification that separates the post period into early and late phases and include only the main DiD and DDD terms. In Panel A, “Southern non-martyr” refers to southern counties not associated with martyr officials, and “Southern martyr” refers to southern counties associated with martyr officials. The equality tests report the p -value for the null that the coefficients for southern non-martyr and southern martyr counties are equal. In Panel B, log distance to Beijing is mean-centered, so the coefficients on “Treat $\times$ Post,” “Treat $\times$ Early,” and “Treat $\times$ Late” are interpretable at the sample mean distance. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: County-Level Stigma vs. Family Retribution

Outcome:	(1) Highest Official Rank	(2) Highest Official Rank	(3) Low Rank (≤ 6)	(4) Low Rank (≤ 6)	(5) Ever Held Central Position	(6) Ever Held Central Position
Same surname $\times$ Post	-0.874** (0.427)		0.159*** (0.055)		-0.136*** (0.047)	
Different surname $\times$ Post	-1.393*** (0.500)		0.215*** (0.057)		-0.104** (0.047)	
Same surname $\times$ Early (1404–1418)		-0.250 (0.474)		0.086 (0.067)		-0.121* (0.062)
Same surname $\times$ Late (1421–1487)		-1.139** (0.469)		0.190*** (0.058)		-0.143*** (0.046)
Different surname $\times$ Early (1404–1418)		-0.637 (0.615)		0.139 (0.093)		-0.102 (0.086)
Different surname $\times$ Late (1421–1487)		-1.629*** (0.538)		0.239*** (0.059)		-0.105** (0.045)
Equality test (p -value)	0.087		0.235		0.223	
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,448	5,448	5,448	5,448	5,448	5,448
Clusters	599	599	599	599	599	599

Notes: “Same surname” indicates *jinshi* from martyr counties sharing a surname with at least one Jingnan martyr. “Different surname” indicates *jinshi* from martyr counties who do not share any martyr surname. Columns (1), (3), and (5) report the static decomposition, and Columns (2), (4), and (6) report the dynamic decomposition, separating the early period (1404–1418) from the late period (1421–1487). Equality test reports the F-test p -value for the Same = Different in the static specification. If the penalty were family retribution, Same should exceed Different; if county stigma, Same and Different should be similar. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Pre-1402 *jinshi* Output and the Martyr-County Penalty

Outcome:	(1) Highest Official Rank	(2) Low Rank (≤ 6)	(3) Ever Held Central Position
Treat $\times$ Post	-6.172*** (1.781)	0.454** (0.218)	-0.258 (0.159)
Treat $\times$ Post $\times$ Log Pre-1402 <i>jinshi</i>	2.645*** (0.861)	-0.156 (0.103)	0.069 (0.075)
Penalty evaluated at:			
No pre-1402 <i>jinshi</i>	-6.172*** (1.781)	0.454** (0.218)	-0.258 (0.159)
Full-sample mean pre-1402 <i>jinshi</i> output	-3.209*** (0.885)	0.279** (0.112)	-0.181** (0.082)
Treated-county mean pre-1402 <i>jinshi</i> output	-1.516*** (0.504)	0.179*** (0.066)	-0.137*** (0.051)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	5,448	5,448	5,448
Clusters	599	599	599

Notes: Log Pre-1402 *jinshi* is measured as $\log(1 + \text{PreJinshi}_c)$, where PreJinshi_c is the number of *jinshi* produced by county c before 1402. The full-sample mean of log pre-jinshi stock is 1.12; the treated-county mean is 1.76. All specifications include Post $\times$ Log Pre-1402 *jinshi*. Standard errors clustered at the county level are in parentheses. Penalty rows report linear combinations evaluated at the indicated stock values. Robustness checks using raw counts, prefecture-level output, and a dynamic split into early (1404–1418) and late (1421–1487) post periods are reported in Table A1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Robustness Checks Summary

Outcome:	(1) Highest Official Rank	(2) Low Rank (≤ 6)	(3) Ever Held Central Position
Panel A: Inference			
Baseline coefficient	-1.039**	0.177***	-0.126***
Wild cluster bootstrap p -value	0.022	0.004	0.007
County randomization inference (p)	0.067	0.014	0.022
Placebo (fake event in 1391–1400)	-0.141	0.049	0.129
Binned pre-trend (p)	0.437	0.403	0.451
Year-by-year pre-trend F-test (p)	0.400	0.619	0.046
Panel B: Alternative Samples & Controls			
Excluding surrender counties	-1.072**	0.177***	-0.131***
Including 1385 cohort	-0.760*	0.090*	-0.108**
South only	-0.937**	0.159***	-0.070*
Matched sample (cutoff = 10)	-2.053***	0.237***	-0.151**
Matched sample (cutoff = 5)	-4.241***	0.436***	-0.194
Matched sample (cutoff = 20)	-1.046**	0.179***	-0.125***
Alternative low rank cutoff (≤ 7)		0.171***	
Alternative low rank cutoff (≤ 8)		0.109*	
+ Exam tier 1/2 indicator (pre-career ability)	-0.963**	0.166***	-0.119***
+ Full controls $\times$ Post (incl. agriculture, population)	-0.796*	0.142**	-0.047
Panel C: Record Missing Bias			
Outcome: Has any record (p)		-0.014 (0.773)	
Outcome: Has any record (including 1385) (p)		-0.043 (0.271)	
90% Range of Simulated Betas	[-1.149, -0.780]	[0.133, 0.190]	[-0.140, -0.088]
Prob. of Remaining Significant ($p < 0.10$)	98.8%	99.9%	98.3%

Notes: This table summarizes robustness checks for the three main career outcomes: Highest Official Rank, Low Rank (≤ 6), and Ever Held Central Position. Panel A reports inference diagnostics and placebo tests. Wild cluster bootstrap uses Rademacher weights with 999 replications. The binned pre-trend test reports the p -value for the null that the pre-period bin coefficient equals zero. Randomization inference is a Fisher-style permutation test based on 1,000 placebo assignments of treated counties. Panel B varies the sample definition and adds alternative controls, including an examination tier 1–2 placement, a proxy for observed examination performance and pre-career ability, geographic controls interacted with Post, and Post interactions with rice suitability, wheat suitability, and county population measured around 1461. Panel C reports selection tests for missing career records and potential misclassification of treatment. Standard errors are not reported in this summary table. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix

A.1 Heterogeneity by county size

A related question is whether the estimated career penalties partly reflect differences in county size rather than place-based political stigma. In the descriptive statistics, treated counties appear larger on average in the available county population data. If larger counties systematically had stronger elite networks or occupied a different position in the Ming administrative hierarchy, county size could shape both baseline career trajectories and the way post-1402 penalties were realized.

Table A3 examines this question by interacting $Treat_c \times Post_t$ with demeaned log county population. Because log population has no substantively meaningful zero, we demean it by subtracting its sample mean. The coefficient on $Treat_c \times Post_t$ is therefore the estimated post-1402 penalty for a county of average population size, while the triple interaction captures how that penalty changes with county population.

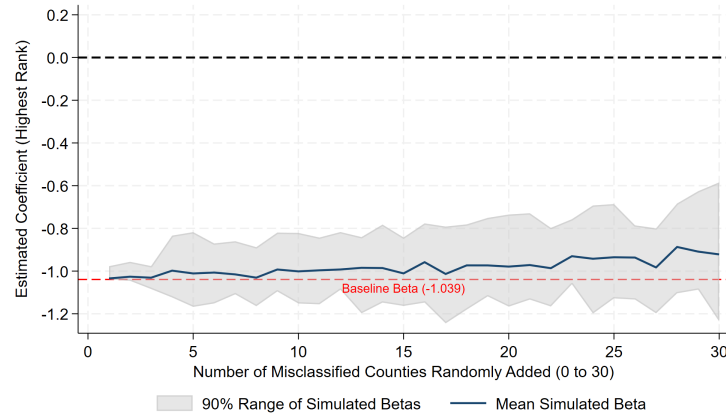
This exercise should be interpreted cautiously, because the available population measure is observed around 1461 (Tianshun era) rather than before the Jingnan conflict, and we therefore use it primarily as a descriptive heterogeneity check rather than a strict pre-treatment robustness test.

The results provide two patterns. First, evaluated at the average county population, the post-1402 penalty remains negative and statistically significant for highest official rank (-2.22 , $p < 0.05$) and positive and statistically significant for low-rank attainment ($+0.20$, $p < 0.05$). For ever holding a central position, the penalty is negative but not statistically distinguishable from zero at the average population size (-0.04 , $p = 0.51$). Second, the moderating effect of population is imprecise. The triple interaction is positive and only marginally significant for highest official rank ($p = 0.076$), and statistically indistinguishable from zero for both low-rank attainment and central appointment. We therefore read these estimates as suggestive descriptive heterogeneity rather than as clean evidence on whether county size shaped the magnitude of the penalty.

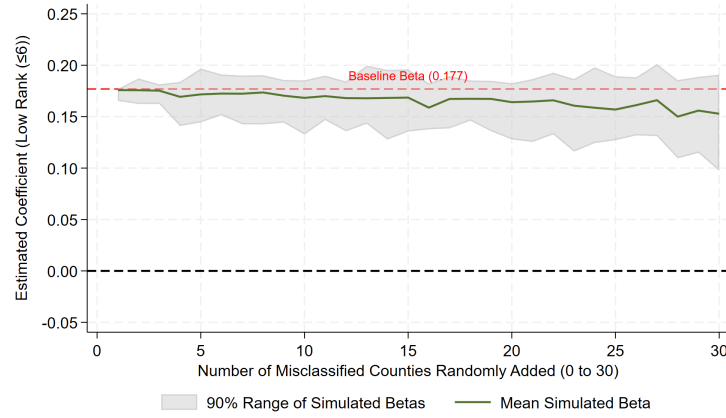
A.2 Additional figures and tables

Figure A.1: Robustness to Potential Unobserved Treated Counties

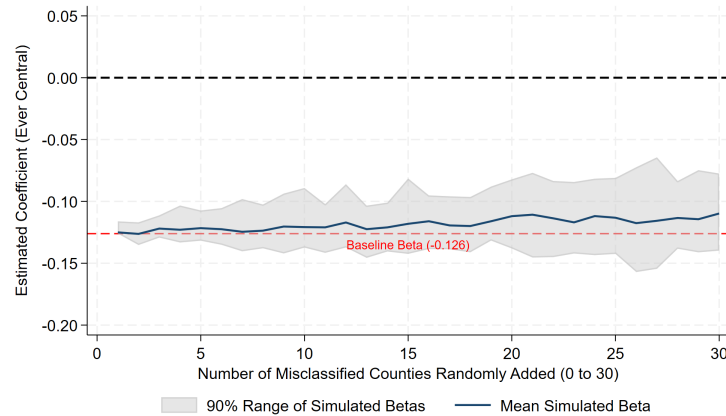
(A) Career Rank



(B) Low Rank Probability



(C) Central Government Access



Note: This figure plots the distribution of re-estimated coefficients across 1,000 permutations. In each permutation, up to 30 pure control counties are randomly reassigned to the treatment group to simulate potential unrecorded treated counties (false negatives). The solid lines represent the mean simulated coefficients, and the shaded areas indicate the 90% confidence bands of the simulated estimates. The horizontal dashed red lines denote the baseline estimates from our main specification, and the black dashed lines mark the zero-effect level.

Table A1: Additional Checks: Pre-1402 *jinshi* Count

Outcome:	(1) Highest Official Rank	(2) Low Rank (≤ 6)	(3) Ever Held Central Position
Panel A: Raw County-Level			
Treat $\times$ Post	-3.231*** (1.024)	0.284** (0.122)	-0.180* (0.100)
Treat $\times$ Post $\times$ Raw Pre-1402 <i>jinshi</i>	0.329** (0.128)	-0.020 (0.015)	0.007 (0.012)
Panel B: Prefecture-Level			
Treat $\times$ Post	-1.505*** (0.493)	0.208*** (0.056)	-0.129** (0.050)
Treat $\times$ Post $\times$ Log Prefecture Pre-1402 <i>jinshi</i>	0.164*** (0.053)	-0.015* (0.008)	0.000 (0.010)
Panel C: County-Level, Dynamic Split			
Treat $\times$ Early	-4.891*** (1.860)	0.310 (0.202)	-0.182 (0.152)
Treat $\times$ Late	-6.580*** (1.850)	0.475** (0.217)	-0.297* (0.158)
Treat $\times$ Early $\times$ Log Pre-1402 <i>jinshi</i>	2.341*** (0.872)	-0.106 (0.111)	0.057 (0.097)
Treat $\times$ Late $\times$ Log Pre-1402 <i>jinshi</i>	2.716*** (0.912)	-0.172 (0.105)	0.073 (0.071)
Equality test: early triple = late triple (<i>p</i> -value)	0.369	0.207	0.793
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	5,448	5,448	5,448
Clusters	599	599	599

Notes: This table reports additional checks for the heterogeneity exercise in Table 7. Panel A replaces the main log county-level pre-*jinshi* measure with the raw number of *jinshi* produced by county *c* before 1402. Panel B measures pre-existing elite pipelines at the prefecture level using the log number of pre-1402 *jinshi* produced by the prefecture. Panel C allows the interaction between the martyr-county penalty and log county-level pre-1402 *jinshi* output to differ between the early post-Jingnan period, 1404–1418, and the later period, 1421–1487. The dynamic specifications include the corresponding lower-order interactions: Early $\times$ Log Pre-1402 *jinshi*, and Late $\times$ Log Pre-1402 *jinshi*. All specifications include county and examination-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Prefecture Fixed Effects with Prefecture-Level Clustering

Outcome:	(1) Tier 1–2	(2) Highest Official Rank	(3) Low Rank (≤ 6)	(4) Ever Held Central Position
Panel A: Static				
Treat $\times$ Post	-0.041 (0.062)	-0.836** (0.394)	0.148*** (0.042)	-0.139*** (0.033)
Treat (level)	0.022 (0.048)	0.554 (0.381)	-0.123*** (0.042)	0.114*** (0.035)
Panel B: Dynamic Split				
Treat $\times$ Early	0.095 (0.105)	-0.311 (0.491)	0.090* (0.050)	-0.149*** (0.055)
Treat $\times$ Late	-0.091* (0.055)	-1.011*** (0.387)	0.168*** (0.045)	-0.135*** (0.035)
Treat (level)	0.024 (0.048)	0.563 (0.378)	-0.124*** (0.043)	0.114*** (0.035)
Prefecture FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	6,916	6,178	6,178	6,178
Clusters (prefectures)	134	133	133	133

Notes: Regressions use prefecture fixed effects and cluster standard errors at the prefecture level. The sample includes individuals whose county is unknown, but prefecture is known (assigned Treat = 0). Standard errors clustered at the prefecture level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Heterogeneity by County Population

Outcome:	(1) Highest Official Rank	(2) Low Rank (≤ 6)	(3) Ever Held Central Position
Treat $\times$ Post	-2.219** (0.944)	0.197** (0.097)	-0.042 (0.063)
Treat $\times$ Post $\times$ Log Population	1.311* (0.737)	-0.031 (0.074)	-0.036 (0.064)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	5,448	5,448	5,448
Clusters	599	599	599

Notes: This table examines whether the post-1402 martyr-county penalty varies with county population size. Log Population is the log of county population measured around 1461, demeaned by subtracting its sample mean. We demean log population because an uncentered value of zero has no substantive interpretation; after demeaning, the coefficient on Treat $\times$ Post is evaluated at the average county population. The coefficient on Treat $\times$ Post $\times$ Log Population shows how the treatment effect changes with a one-unit increase in log county population. All specifications include county and examination-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.